

CONSUMER SEARCH AND PRICE COMPETITION

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CONSUMER SEARCH AND PRICE COMPETITION

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We consider an oligopoly model in which consumers engage in sequential search based on partial product information and advertised prices. By applying Weitzman's (1979) optimal sequential search solution, we derive a simple static condition that fully summarizes consumers' shopping outcomes and translates the pricing game among the sellers into a familiar discrete-choice problem. Exploiting the discrete-choice reformulation, we provide sufficient conditions that guarantee the existence and uniqueness of market equilibrium and analyze the effects of preference diversity and search frictions on market prices. Among other things, we show that a reduction in search costs raises market prices.

KEYWORDS: Consumer search, price advertisements, online shopping, Bertrand competition, product differentiation.

1. INTRODUCTION

WE PRESENT AN OLIGOPOLY MODEL in which consumers sequentially search for the best product based on partial product information and advertised prices. A key distinguishing feature from traditional consumer search models is the observability of prices before search. Consumers still face a nontrivial search problem, because they do not possess full information about their values for the products. Specifically, we consider an oligopoly environment in which a consumer's payoff from purchasing product i is given by $v_i + z_i - p_i$, where v_i (prior value) and p_i (price) are known before search, while she can learn about z_i (match value) only by incurring search costs s_i . In this environment, prices affect each seller's demand not only through their effects on consumers' final purchase decisions, but also through their effects on consumer search behavior. We study how the presence of the latter channel affects sellers' pricing incentives and what its economic consequences are.

Our investigation is mainly motivated by some dramatic changes in retail markets due to the rapid growth of the Internet. The Internet has significantly lowered the cost of collecting price information. Now it is common to check prices online and visit stores

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only to get hands-on information and/or finalize a purchase.¹ This change in consumer behavior has various economic implications. Classic sales tactics that exploit consumers' lack of price information are likely to become obsolete, which will force firms to develop new pricing and advertising strategies. Accordingly, it will become more important in consumer protection to ensure that firms provide enough and accurate information not only at stores, but also on the Internet. Our model can serve as a basic framework to address these and related issues.

In addition, electronic commerce (i.e., online shopping) has already come into our daily lives and is expected to play an increasing role in the economy.² Naturally, the literature on electronic commerce is growing fast in various directions. This paper makes a potentially significant contribution to the literature, because our model captures some salient features of online marketplaces and price comparison websites and, therefore, can yield meaningful insights about how they work. Online shopping typically unfolds as follows: a consumer begins by searching with a specific keyword or choosing an appropriate category. She then faces a summary webpage that displays multiple items with price and brief product descriptions. She clicks a certain set of items, collects more detailed information, and makes a final purchase decision. Our model describes this consumer behavior particularly well.

Price-directed consumer search has been studied by three recent papers, Armstrong and Zhou (2011), Shen (2015), and Haan, Moraga-González, and Petrikaite (2017). Whereas these papers produce several novel insights, each of them confines its analysis to a symmetric duopoly environment,³ mainly because of the complexity of consumer search behavior and resulting complication of demand analysis. Even with two sellers, there are three different paths through which consumers purchase from each seller: some consumers purchase immediately from their first visit. The others visit both sellers and purchase either from their first visit or from their second visit. The number of purchase paths increases factorially fast as the number of sellers increases.⁴

We advance this literature by studying a more general market environment. Specifically, we allow for any number of sellers and asymmetric sellers. This generalization owes much to a new technical result, which we refer to as the eventual purchase theorem.⁵ Each consumer's optimal search strategy can be fully described by an elegant solution by Weitzman (1979), which consists of the optimal search order (in which order to visit the sellers) and the optimal stopping rule (when to purchase or take an outside option). For our purpose, however, it suffices to know optimal search *outcome*, that is, which product each consumer eventually purchases. We show that each consumer's eventual purchase decision can be fully inferred from a static discrete-choice condition with appropriately

¹According to "The 2011 Social Shopping Study" by *PowerReviews*, 57% of respondents compared prices through Google Shopping. In addition, while shopping in physical stores, 36% of consumers checked Amazon prices and 36% looked for other retailers' prices.

²According to *Internet Retailer*, in the United States, total e-commerce sales were \$349.25 billion in 2015 and grew by 14.4% in 2016. Moreover, they accounted for 41.6% of all retail sales growth in 2016.

³The three papers consider different correlation structures for consumers' prior and match values. Both prior and match values are negatively correlated between the products in Armstrong and Zhou (2011), whereas both are independent in Haan, Moraga-González, and Petrikaite (2017). Shen (2015) examined an intermediate case where each consumer's prior values are negatively correlated, while her match values are independent, between the two products. Our model adopts the same independence structure as Haan, Moraga-González, and Petrikaite (2017).

⁴If there are n sellers, then the number of purchase paths for each seller is given by $(n-1)! \sum_{k=0}^{n-1} (n-k)/k!$.

⁵We discuss three relevant studies, Armstrong and Vickers (2015), Armstrong (2017), and Kleinberg, Wagoner, and Weyl (2017) in Section 2.

defined values based on her prior and match values and search costs. The condition reveals that the distortions imposed by search frictions on consumers' purchase decisions take a simple and systematic form.

The eventual purchase theorem allows us to reformulate the pricing game among the sellers as a discrete-choice problem. The reformulation is useful by itself, because it annihilates the need to keep track of different purchase paths. More importantly, it enables us to apply rich existing results on discrete-choice models to our search model. To begin with, it is infeasible to establish the existence of market equilibrium (let alone its uniqueness) with an elementary method: the sellers' best response functions are hard to derive and, even if they can be derived, do not behave well in general (see Haan, Moraga-González, and Petrikaite (2017)). However, based on a general result in the discrete-choice literature, we provide sufficient conditions under which there exists a unique market equilibrium, which is in pure strategies.

We note that sufficient pre-search preference diversity (i.e., consumer heterogeneity) is crucial for the existence of a pure-strategy market equilibrium. If consumers have similar pre-search preferences, then each seller has a strong incentive to undercut the other sellers, because it would induce most consumers to visit her first. This, however, does not lead to marginal-cost pricing in our model, because of ex post preference diversity. In such a case, there can exist only a mixed-strategy equilibrium, whose characterization is exceedingly challenging (see Section C in the Supplemental Material (Choi, Dai, and Kim (2018))). Sufficient ex ante heterogeneity induces consumers to adopt dispersed search strategies, thereby softening price competition and ultimately ensuring the existence of a pure-strategy equilibrium.

We provide a comparative statics result, which is not only useful in our search model, but also contributes to the general discrete-choice literature. The result, whose slightly weaker version has been independently discovered by Zhou (2017), is concerned with a systematic relationship between preference diversity (product differentiation) and equilibrium prices. Within the random utility framework of Perloff and Salop (1985), it has remained an open question what is an appropriate measure of preference diversity, that is, what distributional changes cause market prices to rise. Our result provides an answer to this question: in the symmetric environment, market prices increase if the value distribution becomes more dispersed in the sense of *dispersive order* and does not decrease too much in the sense of first-order stochastic dominance. Importantly, if there is no consumer outside option, as assumed in Perloff and Salop (1985) and many subsequent studies, then the second requirement becomes vacuous and, therefore, dispersive order alone dictates market prices.

As a methodological contribution, we develop an indirect approach to comparative statics regarding search frictions, based on our discrete-choice reformulation and the above general comparative statics result. In our model, the effects of various changes in search frictions can be summarized by their effects on dispersion of the induced discrete-choice distributions. Combining this with our general comparative statics result, we can indirectly learn about the effects of search frictions on market prices, without analyzing their effects on different purchase paths. This allows us to obtain a set of results that are otherwise difficult to obtain in our general environment.

We show that in the symmetric environment, market prices fall with search costs, which is opposite to the classical result on consumer search.⁶ As the value of search decreases,

⁶All three most related papers, Armstrong and Zhou (2011), Shen (2015), and Haan, Moraga-González, and Petrikaite (2017), obtain this result in a duopoly environment. As elaborated in Section 6.2, our contribution

each consumer is less likely to leave for another seller and, therefore, more likely to purchase from the current seller. Each seller then has an incentive to extract more surplus from visiting consumers and, therefore, charges a higher price. This is the main mechanism behind the opposite result in the literature. However, it crucially depends on the assumption of unobservable prices, which implies that the sellers cannot influence consumer search behavior. In our model, the sellers compete in prices to attract consumers. When the value of search falls, it becomes more important for the sellers to attract consumers in the first place, which intensifies price competition and leads to lower prices.

In contrast, improving pre-search information quality has an ambiguous effect on market prices. We show that in a symmetric Gaussian environment, providing more precise product information before consumer search increases market prices if and only if the number of sellers is above a certain threshold. There are two opposing effects. On the one hand, it reduces consumers' incentives to explore more products, which, as above, intensifies price competition among the sellers. On the other hand, consumers' preferences before search (prior values) become more dispersed, which relaxes price competition. We demonstrate that the latter effect dominates the former and, therefore, providing more product information before search increases market prices if there are sufficiently many sellers.⁷

The rest of the paper is organized as follows. We explain how our work is related to various strands of literature in Section 2 and introduce the formal model in Section 3. We analyze consumers' optimal shopping problems in Section 4 and characterize market equilibrium in Section 5. We study the effects of preference diversity and search frictions on market prices in the symmetric environment in Section 6 and conclude in Section 7. All omitted proofs are in the Appendix.⁸

2. RELATED LITERATURE

This paper joins a growing literature on ordered search, which investigates the effects of search order on market outcomes and various ways sellers or intermediaries influence consumer search behavior (order). See Armstrong (2017) for a comprehensive and organized introduction of the literature and several useful discussions. In light of this literature, we consider the case where each consumer's search order is fully endogenized and each seller influences search behavior through his price, which is arguably the most basic instrument.

is to show that this result holds in a more general environment by adopting the indirect approach explained above. As argued and explained in more detail by Haan, Moraga-González, and Petrikaite (2017), this result is consistent with some recent empirical findings. For example, Moraga-González, Sándor, and Wildenbeest (2015) structurally estimated a consumer search model using data from the Dutch automobile market, where price information is believed to be easily accessible, and found that reducing inspection costs led to higher prices for some car models. See also Koulayev (2014), Pires (2017), and Dubois and Perrone (2015) for similar and related findings in different markets.

⁷This result is related to Anderson and Renault (2000). They introduced informed consumers (who know their match values before search) into the model with unobservable prices and showed that equilibrium prices increase in the proportion of informed consumers. It is easy to show that the same result holds in our model with observable prices, but it does not hold more generally: if informed consumers are better informed but still not fully informed about their values, then equilibrium prices may fall in the proportion of informed consumers.

⁸In an earlier version, we provided two sets of results for the case of asymmetric sellers. The main economic messages were (i) that Weitzman values do not provide enough guidance about price rankings in general (i.e., a seller with a lower Weitzman value may announce a higher price), and (ii) that price dispersion (difference) may reduce and a lower cost seller's profit can increase as search costs rise.

Our eventual purchase theorem was anticipated by Armstrong and Vickers (2015) and has been independently discovered by Armstrong (2017) and Kleinberg, Waggoner, and Weyl (2017). Armstrong and Vickers (2015) provided a necessary and sufficient condition for a multiproduct demand system to have a discrete-choice micro-foundation and showed that the condition holds in a consumer search model in which prices are observable and visiting seller 1 is costless, while visiting seller 2 is costly. Our theorem can be interpreted as an application of their general result, and a closed-form solution, to a more general environment. Armstrong (2017) derived an effectively identical condition to ours but used the result to motivate and discuss various ideas on ordered search, rather than exploiting it to study a specific consumer search model. Kleinberg, Waggoner, and Weyl (2017) also obtained a closely related result. They considered an auction environment in which each bidder is required to inspect and learn about his full value in order to acquire a good and showed that the Dutch (descending-price) auction often outperforms other common auctions in such an environment. Their analysis utilizes a new “real options” interpretation of Weitzman’s solution. Although they did not report a result on eventual choice, our eventual purchase theorem can be inferred from their results.

Dispersive order, which we utilize in order to establish comparative statics regarding search frictions, has been found useful in several other economic applications. Zhou (2017) independently discovered an almost identical result to our dispersive order result and used it to study the effects of bundling in the Perloff–Salop framework. Our result is slightly more general than his, because we allow for consumer outside option. Ganuza and Penalva (2010) showed that dispersive order is useful in evaluating the effects of information disclosure in symmetric private-value second-price auctions: as the distribution of conditional expected values becomes more dispersed, efficiency always improves, but seller surplus does not necessarily increase. Bhaskar and Hopkins (2016) considered a frictionless two-sided matching market with noisy premarital investments and showed that dispersive order can be used to predict which side over- or under-invest relative to the efficient level.

As explained in Section 1, one of the most important economic implications of observable prices is that market prices decrease with search costs. Several recent papers identify other mechanisms that yield the same result. For example, Zhou (2014) considered a multiproduct search model and showed that the joint search effect (that decreasing one product’s price raises the other product’s demand as well) can outweigh the usual market power effect and, therefore, prices can decrease with search costs. Moraga-González, Sándor, and Wildenbeest (2017) introduced heterogeneous search costs and consumer outside option into the model of Wolinsky (1986) and demonstrated that the extensive search margin (how many consumers choose to search in the first place) may induce prices to decrease as search costs increase in the sense of first- or second-order stochastic dominance. See also Garcia, Honda, and Janssen (2017), who studied a model in which both consumers and retailers search within a vertical industry structure, and Shelegia and Garcia (2018), who analyzed an observational learning model in which each consumer observes the purchase decision of the immediate predecessor before search.

There is a fairly large literature on directed search, which has been developed mainly in the labor context: see a recent comprehensive survey by Wright, Kircher, Julien, and Guerrieri (2017). The main economic problem in those models is congestion among searchers (i.e., coordination frictions) resulting from capacity constraints (each firm has one vacancy, or each seller has unit supply), which is absent in our model. In addition, in most directed search models, each searcher faces a stationary or static search problem, which is considerably simpler than a consumer’s non-stationary sequential problem in our model.

Our paper is also related to two strands of literature on electronic commerce. First, several papers have developed an equilibrium online shopping model. For example, Baye and Morgan (2001) presented a model in which both sellers and consumers decide whether to participate in an online marketplace, while Chen and He (2011), Athey and Ellison (2011), and Anderson and Renault (2017) presented a model that combines position auctions with consumer search. Our paper is unique in that the focus is on consumer search within an online marketplace. Second, a growing number of papers draw on search theory to study online markets. For example, Kim, Albuquerque, and Bronnenberg (2010) developed a non-stationary search model to study the online market for camcoders. De los Santos, Hortaçsu, and Wildenbeest (2012) tested some classical search theories with online book sale data and argued that simultaneous search explains the data better than sequential search. Dinerstein, Einav, Levin, and Sundaresan (2017) estimated online search costs and retail margins with a consumer search model based on the “consideration set” approach, and applied them to evaluate the effects of a search redesign by eBay in 2011. Although empirical analysis is beyond the scope of this paper, we think that our equilibrium model is tractable and structured enough to be taken to data.

3. ENVIRONMENT

The market consists of n sellers, each indexed by $i = \{1, \dots, n\}$, and a unit mass of consumers. The sellers face no capacity constraint, while each consumer demands one unit among all products. The sellers simultaneously announce prices. Consumers observe those prices and sequentially search for the best product.

Each seller i supplies a product at no fixed cost and constant marginal cost c_i . We denote by $p_i \in \mathcal{R}_+$ seller i 's price. In addition, we let $\mathbf{p}$ denote the price vector for all sellers (i.e., $\mathbf{p} = (p_1, \dots, p_n)$) and $\mathbf{p}_{-i}$ denote the price vector except for seller i 's price (i.e., $\mathbf{p}_{-i} = (p_1, \dots, p_{i-1}, p_{i+1}, \dots, p_n)$). Denote by $D_i(\mathbf{p})$ the measure of consumers who eventually purchase product i . Seller i 's profit is then defined to be $\pi_i(\mathbf{p}) \equiv D_i(\mathbf{p})(p_i - c_i)$. Each seller maximizes his profit $\pi_i(\mathbf{p})$.

Each consumer's random utility for seller i 's product is given by $\widehat{V}_i = V_i + Z_i$. The first component V_i is the (representative) consumer's prior value for product i , while the second component Z_i is the residual part that is revealed to the consumer only when she visits seller i and inspects his product. As for prices, we let $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{z} = (z_1, \dots, z_n)$ denote realized value profiles for each component.

The products are horizontally differentiated. Specifically, for each consumer, V_i and Z_i are independently and identically drawn from the intervals $[\underline{v}_i, \bar{v}_i]$ and $[\underline{z}_i, \bar{z}_i]$ according to the distribution functions F_i and G_i , respectively. In addition, they are independent of each other and across the products. We allow each support to be infinite on both sides and assume that F_i and G_i have continuously differentiable density functions f_i and g_i , respectively. Independence across the products enables us to utilize the optimal search solution by Weitzman (1979), while independence between V_i and Z_i yields a clean and easy-to-interpret characterization.⁹

⁹Independence between V_i and Z_i is restrictive not by itself, but because of a joint additive-utility specification ($\widehat{V}_i = V_i + Z_i$). We note that our specification generalizes Haan, Moraga-González, and Petrikaite (2017), but does not encompass Armstrong and Zhou (2011) and Shen (2015) (see footnote 3). As well-recognized in the literature (e.g., Perloff and Salop (1985), Quint (2014), Zhou (2014)), one important advantage of our specification is that it can readily accommodate any number of sellers. It also makes our results directly comparable to the standard results with unobservable prices (Wolinsky (1986), Anderson and Renault (1999)).

Search is costly, but recall is costless. Specifically, each consumer must visit seller i and discover her match value z_i in order to be able to purchase product i .¹⁰ She incurs search costs $s_i (> 0)$ on her first visit. Then, she can purchase the product immediately or recall it any time later. Each consumer can leave the market and take an outside option at any point. The value of the outside option is given by u_0 for all consumers.

Each consumer's ex post utility depends on her total value for the purchased product $\hat{v}_i$, its price p_i , and her search history. Let N be the set of sellers the consumer visits. If she purchases product i (in N), then her ex post utility is equal to

$$U(v_i, z_i, p_i, N) = v_i + z_i - p_i - \sum_{j \in N} s_j.$$

If she takes the outside option, then her ex post utility is equal to $U(N) = u_0 - \sum_{j \in N} s_j$. Each consumer is risk neutral and maximizes her expected utility.

The market proceeds as follows. First, the sellers simultaneously announce prices $\mathbf{p}$. Then, each consumer shops (searches) based on available information $(\mathbf{p}, \mathbf{v})$. We study subgame perfect Nash equilibria of this market game.¹¹ We first characterize consumers' optimal shopping behavior and then analyze the pricing game among the sellers.

4. CONSUMER BEHAVIOR

In this section, we analyze consumers' sequential search problems.

4.1. Optimal Shopping

Given prices $\mathbf{p}$ and prior values $\mathbf{v}$, each consumer faces a sequential search problem. She decides in which order to visit the sellers and, after each visit, whether to stop, in which case she chooses which product to purchase, if any, among those she has inspected so far, or visit another seller. Although this is a complex combinatorial problem in general, an elegant solution is available by Weitzman (1979). Independence between V_i and Z_i leads to an even sharper characterization, as reported in the following proposition.¹²

PROPOSITION 1: *Given $\mathbf{p} = (p_1, \dots, p_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$, the (representative) consumer's optimal search strategy is as follows: for each i , let z_i^* be the value such that*

$$s_i = \int_{z_i^*}^{\bar{z}_i} (1 - G_i(z_i)) dz_i. \quad (1)$$

(i) *Search order: the consumer visits the sellers in the descending order of $v_i + z_i^* - p_i$.*

¹⁰We do not allow the possibility that a consumer purchases a product without inspecting it. Although the possibility arises naturally in some contexts, it significantly complicates the sequential search problem and does not permit a general and tractable solution. In particular, Weitzman's solution applies unchanged only in some special cases (Doval (2018)).

¹¹For notational simplicity, we do not formally define consumers' search strategies. See Weitzman (1979) for a formal recursive description of search strategy.

¹²The measure of consumers who are indifferent over multiple choices is negligible, because F_i and G_i are continuously increasing for all i . We omit a description of those consumers' behavior.

(ii) *Stopping*: let N be the set of sellers the consumer has visited so far. She stops, and takes the best available option by the point, if and only if

$$\max\left\{u_0, \max_{i \in N} v_i + z_i - p_i\right\} > \max_{j \notin N} v_j + z_j^* - p_j.$$

Weitzman's solution is based on a single index for each option (seller). Let r_i be the value ("reservation utility") such that the consumer is indifferent between obtaining utility r_i immediately and visiting seller i :

$$r_i = -s_i + \int_{z_i}^{\bar{z}_i} \max\{r_i, v_i + z_i - p_i\} dG_i(z_i).$$

Weitzman (1979) showed that the optimal search strategy is to visit the sellers in the decreasing order of r_i and stop as soon as the best realized utility by the point exceeds all remaining r_i 's. In our model, due to the random additive-utility specification, $r_i = v_i + z_i^* - p_i$, where z_i^* is given by equation (1). Notice that z_i^* is independent of v_i and p_i but strictly decreasing in s_i . Intuitively, z_i^* represents the net option value of visiting seller i and learning z_i . Therefore, it inherits the independence properties of Z_i and decreases as search costs s_i rise.

4.2. Shopping Outcomes

Despite the simplicity of Weitzman's solution, it is nontrivial to summarize consumers' shopping outcomes and compute demand for each seller. The main difficulty lies in that demand for each seller consists of several different purchase paths, whose number increases factorially fast in the number of sellers n . We provide a simple way to circumvent this difficulty. The following result shows that, whereas Weitzman's solution is necessary to fully describe a consumer's optimal purchase path, her eventual purchase decision can be fully summarized by a simple static condition.

THEOREM 1—Eventual Purchase: *Let $w_i \equiv v_i + \min\{z_i, z_i^*\}$ for each i . Given $(\mathbf{p}, \mathbf{v}, \mathbf{z})$, the consumer purchases product i if and only if $w_i - p_i > u_0$ and $w_i - p_i > w_j - p_j$ for all $j \neq i$.*

PROOF: Sufficiency: $w_i - p_i > u_0$ implies that the consumer purchases a product, because she is willing to visit at least one seller ($v_i + z_i^* - p_i > u_0$) and make a purchase ($v_i + z_i - p_i > u_0$). Given this, it suffices to show that if $w_i - p_i > w_j - p_j$, then she never purchases product j .

- Suppose $z_j^* \leq z_j$, which implies that $w_j = v_j + z_j^*$. The consumer visits seller j only after seller i because $v_i + z_i^* - p_i \geq w_i - p_i > v_j + z_j^* - p_j$. However, once she visits seller i , she has no incentive to visit seller j because $v_i + z_i - p_i > v_j + z_j^* - p_j$.

- Suppose $z_j^* > z_j$, which implies that $w_j = v_j + z_j$. In this case, even if she visits seller j , she either recalls a previous product ($v_i + z_i - p_i > v_j + z_j - p_j$) or continues to search ($v_i + z_i^* - p_i > v_j + z_j - p_j$) and finds a better product ($v_i + z_i - p_i > v_j + z_j - p_j$).

Necessity: if $w_i - p_i < u_0$, then the consumer does not visit seller i ($v_i + z_i^* - p_i < u_0$) or does not purchase product i even if she visits seller i ($v_i + z_i - p_i < u_0$). If $w_i - p_i < w_j - p_j$ for some $j \neq i$, then, for the same logic as above, the consumer never purchases product i . Q.E.D.

Theorem 1 suggests that each consumer’s eventual purchase decision can be represented as in canonical discrete-choice models.¹³ The only difference is that consumers’ purchase decisions are made based, neither on true values $\widehat{v}_i$ nor on prior values v_i , but on newly identified values w_i , which we refer to as *effective values*. Clearly, w_i is related to underlying values $\widehat{v}_i$ and v_i . In particular, w_i converges to $\widehat{v}_i$ as s_i tends to 0 (in which case z_i^* approaches $\bar{z}_i$) and is determined only by v_i as s_i tends to infinity (in which case z_i^* approaches $-\infty$). Intuitively, if there are no search costs, each consumer makes a fully informed decision and purchases the product that offers the largest net utility (i.e., $w_i = \widehat{v}_i$ for all i). To the contrary, if search costs grow arbitrarily large, then consumers’ purchase decisions depend only on prior values. In general, search frictions make consumers’ match values $\mathbf{z}$ imperfectly reflected in their purchase decisions. The problem becomes more severe, and consumers rely less on $\mathbf{z}$, as search frictions increase.

The specific upper truncation structure of w_i derives from the monotonicity properties of Weitzman’s solution. If a consumer visits seller i , that means that she has not found a product that gives her more than $v_i + z_i^* - p_i$ (optimal stopping rule) and Weitzman’s indices for all remaining sellers are lower than $v_i + z_i^* - p_i$ (optimal search rule). Therefore, she stops and purchases product i immediately whenever $z_i \geq z_i^*$. This implies that each consumer’s eventual purchase probability is independent of z_i conditional on $z_i > z_i^*$, which is equivalent to z_i ’s entering into consumers’ purchase decisions only through $\min\{z_i, z_i^*\}$.

Theorem 1 provides a particularly simple way to derive demand for each seller, as elaborated in the next section. It also significantly simplifies the calculation of consumer surplus, as shown in the following result.¹⁴

COROLLARY 1: *Given $\mathbf{p}$, the expected payoff of a consumer with prior values $\mathbf{v}$ is equal to $E[\max\{u_0, \max_i v_i + \min\{Z_i, z_i^*\} - p_i\}]$.*

Corollary 1 states that consumer surplus can also be calculated as in discrete-choice models: a consumer’s expected payoff is equal to the expectation of her highest effective value. It is noteworthy that consumer surplus does not directly depend on search costs s_i : the effects of search costs are fully reflected in z_i^* and, therefore, they influence consumer surplus only through effective values $w_i = v_i + \min\{z_i, z_i^*\}$.

5. MARKET EQUILIBRIUM

In order to utilize Theorem 1, let H_i denote the distribution function for the new random variable $W_i = V_i + \min\{Z_i, z_i^*\}$, that is,¹⁵

$$H_i(w_i) \equiv \int_{z_i}^{z_i^*} F_i(w_i - z_i) dG_i(z_i) + \int_{z_i^*}^{\bar{z}_i} F_i(w_i - z_i^*) dG_i(z_i). \quad (2)$$

¹³Theorem 1 holds even if prices are not observable to consumers before search, as long as consumers have correct beliefs about prices (i.e., in equilibrium). However, the result does not hold if a seller deviates, because consumers’ search decisions are based on their expectations about prices, while their final purchase decisions depend on actual prices charged. Still, the condition can be modified in a straightforward fashion and used to study the model with unobservable prices. See Section 6.2.

¹⁴A similar result also appears in Armstrong (2017) and Kleinberg, Waggoner, and Weyl (2017).

¹⁵We let $\underline{w}_i \equiv \underline{v}_i + \min\{z_i, z_i^*\}$ and $\bar{w}_i \equiv \bar{v}_i + z_i^*$, so that $[\underline{w}_i, \bar{w}_i]$ is the support of H_i . A closed-form solution for $H_i(w_i)$ is available for some commonly used distributions. See the Supplemental Material (Section A) for three examples.

In addition, let $X_i \equiv \max\{u_0, \max_{j \neq i} W_j - p_j\}$ denote each consumer's (random) utility from the best alternative to product i and $\tilde{H}_i(x_i) \equiv \Pr\{X_i \leq x_i\}$ denote its distribution. Since each consumer purchases product i if and only if $w_i - p_i$ exceeds u_0 and $w_j - p_j$ for all $j \neq i$, aggregate demand for product i is equal to

$$D_i(\mathbf{p}) = \int (1 - H_i(x_i + p_i)) d\tilde{H}_i(x_i).$$

From the above discrete-choice representation, it is easy to derive the following necessary (first-order) condition for equilibrium p_i :

$$\frac{1}{p_i - c_i} = -\frac{dD_i(\mathbf{p})/dp_i}{D_i(\mathbf{p})} = \frac{\int h_i(x_i + p_i) d\tilde{H}_i(x_i)}{\int (1 - H_i(x_i + p_i)) d\tilde{H}_i(x_i)}. \quad (3)$$

This is a standard optimal pricing formula: $-dD_i(\mathbf{p})/dp_i$ is the measure of marginal consumers who are indifferent between product i and the best alternative (i.e., $w_i - p_i = x_i$). The formula states that the optimal markup is inversely proportional to the proportion of marginal consumers among those who purchase product i .

The demand system above exhibits standard properties for imperfect substitutes: demand for seller i decreases in own price p_i and increases in competitors' prices $\mathbf{p}_{-i}$. However, it does not behave well in general: depending on the precise shapes of H_i and $\tilde{H}_i$, the right-hand side in equation (3) may not be monotone in $\mathbf{p}$, in which case there may exist no or multiple pure-strategy equilibria. Our discrete-choice reformulation, however, allows us to borrow existing results on discrete-choice models, rather than developing our own specific results. This is particularly useful for equilibrium existence and uniqueness, for which various general results are available.

We establish the existence and uniqueness of market equilibrium by building upon Quint (2014), who provided one of the most general characterizations for standard discrete-choice models. His result makes use of the following two assumptions:

ASSUMPTION 1: For each i , both H_i and $1 - H_i$ are log-concave.

ASSUMPTION 2: For each i , the support of H_i has no upper bound (i.e., $\bar{w}_i = \infty$).

The following result is a special version of his Theorem 1 and Lemma 1.

THEOREM 2—Quint (2014): Under Assumption 1, $D_i(\mathbf{p})$ is log-concave in p_i , and $\log D_i(\mathbf{p})$ has strictly increasing differences in p_i and p_j . Under Assumptions 1 and 2, there exists a unique equilibrium, which is in pure strategies, in the pricing game among the sellers.

Intuitively, log-concavity of $1 - H_i$ (i.e., $h_i/(1 - H_i)$ increasing) generates the effect of pushing up the right-hand side in equation (3) as p_i rises. Log-concavity of $\tilde{H}_i$ (which stems from log-concavity of H_j 's) reinforces the basic effect, thereby ensuring not only that seller i 's best response to $\mathbf{p}_{-i}$ is unique, but also that the pricing game is a supermodular game. Assumption 2 guarantees that each seller's demand is positive at any price and, therefore, each seller always chooses a price above c_i . The existence of a pure-strategy equilibrium follows as an application of more general existence theorems for supermodular games (see, e.g., Vives (2005)). Its uniqueness is not implied by general theory but can

be obtained by exploiting the linear structure of the model, namely, that $D_i(\mathbf{p})$ is invariant when all prices, together with $-u_0$, increase by the same amount.

Unlike in standard discrete-choice models where H_i 's are primitives, they depend on F_i , G_i , and s_i in a specific way in our model, as shown in equation (2). Indeed, Assumption 1 can fail even with a strong assumption that both f_i and g_i are log-concave.¹⁶ In order to understand the underlying problem, consider the case where F_i is degenerate at v_i . In this case, $H_i(w_i)$ jumps up at $v_i + z_i^*$ (see the solid line, corresponding to $\alpha = 0$, in the left panel of Figure 1) and, therefore, cannot be globally log-concave. When F_i is not degenerate, the atom at $v_i + z_i^*$ is continuously scattered, in which case H_i is continuous. However, if F_i is sufficiently concentrated around v_i , then the slope of H_i at $v_i + z_i^*$ can be arbitrarily large (see the dashed line, corresponding to $\alpha = 0.1$, in the left panel of Figure 1). Therefore, H_i may still fail to be log-concave.

Based on this observation, we provide a set of sufficient conditions under which Theorem 2 applies to our model.¹⁷

PROPOSITION 2: *Suppose that both f_i and g_i are log-concave and the support of F_i has no upper bound (which is necessary and sufficient for Assumption 2).*

- *The survival function $1 - H_i$ is always log-concave.*

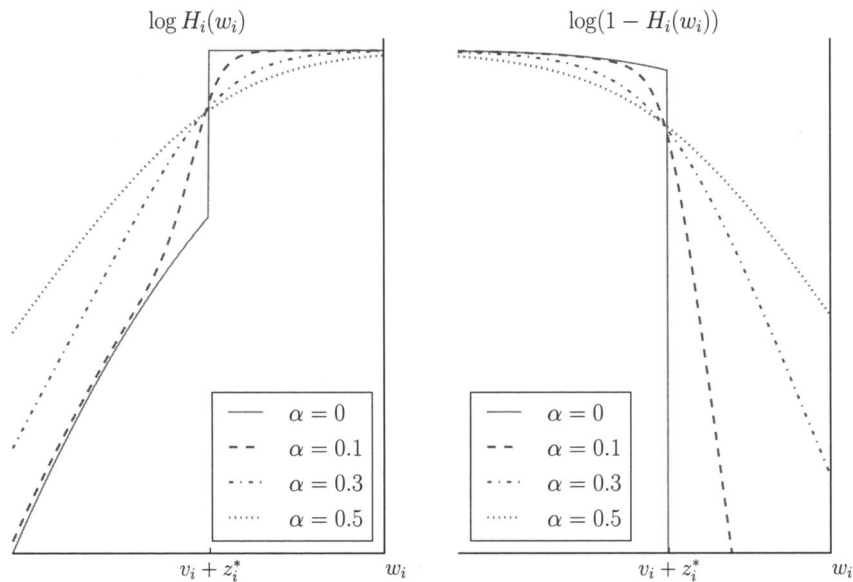


FIGURE 1.— $\log(H_i(w_i))$ and $\log(1 - H_i(w_i))$ for different dispersion levels of F_i . For both panels, $F_i(v_i) = 1/(1 + e^{-v_i/\alpha})$ (logistic), $G_i(z_i) = \Phi(z_i)$ (standard normal), and $s_i = 1$.

¹⁶If the density function f is log-concave, then both distribution function F and survival function $1 - F$ are log-concave by Prékopa's theorem. See Bagnoli and Bergstrom (2005) for more details.

¹⁷Note that Assumptions 1 and 2 are sufficient but not necessary for the existence of a pure-strategy equilibrium and, therefore, their violation does not imply that there does not exist a pure-strategy equilibrium. Although it is beyond the scope of this paper to provide a more general sufficient condition, it is easy to see that there cannot exist a pure-strategy equilibrium if F_i 's are degenerate (see the explanation in Section 1).

- If the variance of V_i is sufficiently large and F_i has no lower bound (i.e., $\underline{v}_i = -\infty$) or $f_i(\underline{v}_i) = 0$, then H_i is also log-concave.¹⁸
- If $f'_i(u_0 + c_i - z_i^*) \leq 0$, then H_i is log-concave above $w_i > \max\{u_0 + c_i, \underline{w}_i\}$.

The first result states that unlike H_i , log-concavity of f_i and g_i suffices for log-concavity of $1 - H_i$. The difference lies in the fact that, as shown in the right panel of Figure 1, when F_i is degenerate at v_i , $1 - H_i$ jumps down at the discontinuity point $v_i + z_i^*$, which does not disrupt log-concavity of the function. Log-concavity of f_i and g_i also guarantees log-concavity of H_i in the limit as s_i tends to 0 (in which case $z_i^* \rightarrow \bar{z}_i$) or ∞ (in which case $z_i^* \rightarrow -\infty$), but more restrictions are required when $s_i \in (0, \infty)$.

The second result is based on the idea that if F_i is sufficiently spread out, then the atom of $\min\{Z_i, z^*\}$ at z^* is so well scattered out through the support that H_i can be log-concave. The requirement on $\underline{v}_i$ ensures that h_i does not jump at $w_i = \underline{v}_i + z_i^*$. In the left panel of Figure 1, if α (which measures dispersion of F_i) is sufficiently large, then H_i increases sufficiently slowly around $v_i + z_i^*$ and, therefore, becomes log-concave.¹⁹ Intuitively, as F_i becomes more spread out, consumers' search and purchase decisions become less sensitive to p_i . This ensures that seller i 's demand varies sufficiently gradually in p_i that she always has a unique and well-behaved best response to $\mathbf{p}_{-i}$.

The final result is based on the observation that Theorem 2 applies as long as Assumption 1 holds for a relevant parameter region. Since each seller always chooses $p_i \geq c_i$, consumers such that $w_i \leq u_0 + c_i$ are effectively irrelevant to seller i 's optimal pricing, because $w_i - p_i \leq w_i - c_i \leq u_0$ and, therefore, those consumers never purchase from the seller. A simple condition $f'_i(u_0 + c_i - z_i^*) \leq 0$ ensures that H_i is log-concave for all relevant consumers.²⁰ This condition always holds with monotone density functions (e.g., exponential and half-normal) and is trivial to check with most distribution functions. In addition, if the support of F_i has no upper bound (as implied by Assumption 2), then it must eventually decrease. Therefore, the condition necessarily holds if $u_0 + c_i$ is sufficiently large. See the Supplemental Material (Section A) for some straightforward examples.

6. COMPARATIVE STATICS

In this section, we present a set of comparative statics results regarding preference diversity and search frictions.²¹ For clear insights as well as tractability, we restrict attention to the case where the sellers are symmetric. Precisely, we assume that for all i , $F_i = F$, $G_i = G$, $c_i = c$, $s_i = s$, and $H_i = H$. We also maintain Assumptions 1 and 2,²² so that there

¹⁸To be precise, we consider $V_i^\sigma = \sigma V_i$ for a given V_i and show that the effective value distribution H_i^σ corresponding to V_i^σ is log-concave if σ is sufficiently large.

¹⁹Haan, Moraga-González, and Petrikaite (2017) conjectured this result and provided a set of confirming numerical examples. Our result formalizes their conjecture. In the example used in Figure 1, numerically, H_i is log-concave when $\alpha \geq 0.281$. If $F_i = \mathcal{N}(0, \alpha^2)$, instead, then H_i is log-concave when $\alpha \geq 0.621$.

²⁰We thank an anonymous referee for suggesting this simple and useful condition.

²¹Given our discrete-choice reformulation, most standard results immediately follow from the existing literature. See Quint (2014) for several basic results. One important question in the literature is the effect of more intense competition (i.e., the number of sellers n) on market prices. For example, Chen and Riordan (2008) provided a condition under which the symmetric duopoly price is higher than the monopoly price, and Gabaix, Laibson, Li, Li, Resnick, and de Vries (2016) analyzed asymptotic price behavior when there are numerous sellers (i.e., as n tends to ∞).

²²To be precise, we require H to be log-concave only above $u_0 + c$. Alternatively, one can directly assume that the symmetric equilibrium price is characterized by the first-order condition and $D_i(\mathbf{p})$ satisfies the properties in Theorem 2.

is a unique equilibrium, which is in symmetric pure strategies, and $D_i(\mathbf{p})$ is well-behaved. We let p^* and $\pi(p^*)$ denote the symmetric equilibrium price and profit, respectively.

6.1. Preference Diversity

We begin by establishing a result that relates dispersion of H to p^* . The result not only is useful for our subsequent comparative statics, but also has a direct contribution to the general discrete-choice literature. Product differentiation provides a way to overcome the Bertrand paradox: each seller has some loyal consumers (who value the seller's product more than other products) and, therefore, can set a positive markup even under Bertrand competition. Given this observation, it is plausible that the more differentiated consumers' preferences are, the higher prices the sellers charge. A challenge has been to identify an appropriate measure of product differentiation. In their seminal work, Perloff and Salop (1985) showed that constant scaling of consumers' preferences necessarily increases market prices, but found that the result does not extend for mean-preserving spreads. Our result offers an answer to this question.

We utilize the following stochastic order, typically referred to as *dispersive order*.

DEFINITION 1: The distribution function H_2 is more dispersed than the distribution function H_1 if $H_2^{-1}(b) - H_2^{-1}(a) \geq H_1^{-1}(b) - H_1^{-1}(a)$ for any $0 < a \leq b < 1$. Likewise, H_2 is more dispersed than H_1 above w if $H_2^{-1}(b) - H_2^{-1}(a) \geq H_1^{-1}(b) - H_1^{-1}(a)$ for any $H_1(w) < a \leq b < 1$.²³

Intuitively, a more dispersed distribution has more spread-out density and, therefore, increases more slowly, which is equivalent to its quantile function H^{-1} increasing faster.²⁴ Dispersive order is useful in various problems (see Section 2) but also has some limitations. In particular, two random variables with a common finite support can never be ranked in terms of dispersive order.²⁵ This problem, however, does not arise in our model because of Assumption 2. The second definition with qualifier "above w " is useful in our model, because the left tail of H does not affect p^* , as explained for the last result of Proposition 2.

PROPOSITION 3: *In the symmetric environment, the equilibrium price p^* increases as H becomes more dispersed above $u_0 + c$ and $H(u_0 + c)$ weakly decreases.*

PROOF: In the symmetric environment, equation (3) reduces to

$$\frac{1}{p^* - c} = \frac{\int_w^\infty h(\max\{u_0 + p^*, w\}) dH(w)^{n-1}}{\int_w^\infty (1 - H(\max\{u_0 + p^*, w\})) dH(w)^{n-1}}. \quad (4)$$

²³Note that the definition requires that the inequality hold for all interior quantiles but imposes no restriction on the boundary points. This fact is used in our proofs of Propositions 4 and 5.

²⁴Dispersive order is location-free and, therefore, neither is implied by nor implies second-order stochastic dominance. Mean-preserving dispersive order, however, implies mean-preserving spread: if H_2 is more dispersed than H_1 with the same mean, then H_2 is a mean-preserving spread of H_1 . See Shaked and Shanthikumar (2007) for further details.

²⁵Suppose H_1 and H_2 have the same bounded support $[\underline{w}, \bar{w}]$. By definition,

$$H_1^{-1}(1) - H_1^{-1}(0) = H_2^{-1}(1) - H_2^{-1}(0) = \bar{w} - \underline{w}.$$

But then, $H_2^{-1}(b) - H_2^{-1}(a) \geq H_1^{-1}(b) - H_1^{-1}(a)$ for any $0 < a \leq b < 1$ if and only if $H_1 = H_2$.

The left-hand side strictly decreases in p^* , while the right-hand side increases in p^* (see the Appendix). Therefore, it suffices to prove that given p^* , the right-hand side falls when H changes as indicated. Let $\phi \equiv H(u_0 + p^*)$ and change the variable with $a = H(w)$. Then,

$$\frac{1}{p^* - c} = \frac{h(H^{-1}(\phi))\phi^{n-1} + \int_{\phi}^1 h(H^{-1}(a)) da^{n-1}}{\frac{1}{n}(1 - \phi^n)}.$$

If H becomes more dispersed above $u_0 + c$, then $dH^{-1}(a)/da = 1/h(H^{-1}(a))$ increases for each $a \geq \phi$, which lowers the right-hand side. If, in addition, $H(u_0 + c)$ decreases, then ϕ also decreases, because $p^* > c$ and a distribution function crosses a less dispersed one only once from above. This further lowers the right-hand side, because it has the same effect as lowering p^* , which enters only through $\phi = H(u_0 + p^*)$, in the right-hand side. Q.E.D.

The relationship between dispersion of H and p^* is particularly clear when there is no outside option (i.e., $u_0 = -\infty$). In that case, the second condition regarding $H(u_0 + c)$ is vacuous and p^* depends only on the dispersive order of H . In order to understand this result, notice that in the symmetric equilibrium with no outside option, each consumer purchases product i if and only if her effective value for product i exceeds those for the other products. This means that $D_i(\mathbf{p})$ is identical to the probability that W_i is the first-order statistic among all W_j 's, and $-dD_i(\mathbf{p})/dp_i$ is identical to the probability density that W_i coincides with the second-order statistic $\max_{j \neq i} W_j$. As H becomes more dispersed, the difference between the first- and the second-order statistics grows larger and, therefore, the density of the event $\{W_i = \max_{j \neq i} W_j\}$ decreases. This implies fewer marginal consumers and, therefore, higher market prices.²⁶

In general, dispersive order alone does not suffice for the result. This is because dispersive order is concerned only with the slope of the quantile function and, therefore, location-free. For example, $\mathcal{N}(\mu_2, \sigma_2^2)$ is more dispersed than $\mathcal{N}(\mu_1, \sigma_1^2)$ if and only if $\sigma_2 > \sigma_1$, independent of μ_1 and μ_2 . When there is no outside option, each consumer must purchase one of the products, even if she receives arbitrarily low negative utility. In that case, the location of H does not affect demand and, therefore, p^* is determined only by dispersion of H . When there is a finite outside option, the location clearly matters: if H shifts to the left, then more consumers would opt to take the outside option, which provides an incentive for the sellers to lower their prices. The second requirement in Proposition 3 ensures that H does not shift to the left so much that this location effect does not disrupt the dispersion effect. Naturally, a sufficient condition for the second requirement is that H increases in the sense of first-order stochastic dominance.²⁷

In our model of consumer search, H is determined by the two primitive distributions F (prior values) and G (match values). The following proposition illustrates how dispersion

²⁶This explanation, based on the relationship between the first- and the second-order statistics, is borrowed from Zhou (2017). See also Gabaix et al. (2016).

²⁷The two conditions in Proposition 3 also hold together in the following two cases: (i) H becomes more dispersed while its mean stays constant (mean-preserving dispersion), and $u_0 + c$ is sufficiently large. If H is symmetric and single-peaked around μ_w , then it suffices that $u_0 + c > \mu_w$. (ii) H has log-concave density, and an independent positive noise ε is added to w . In this case, $w + \varepsilon$ is more dispersive and stochastically higher than w (Theorem 3.B.7 in Shaked and Shanthikumar (2007)).

of each distribution translates into dispersion of H and, therefore, affects market prices (combined with Proposition 3).

PROPOSITION 4: *In the symmetric environment, H becomes more dispersed above $u_0 + c$*

- *as G becomes more dispersed, provided that f is log-concave, or*
- *as F becomes more dispersed, provided that g is log-concave, F has decreasing density above $u_0 + c - z^*$, and $\underline{v} \in (-\infty, u_0 + c - z^*]$.*

PROOF: We use the following mathematical result: if X_1 has log-concave density, then $X \equiv X_1 + X_2$ becomes more dispersed whenever X_2 becomes more dispersed (Theorem 3.B.8 in Shaked and Shanthikumar (2007)). The first result directly follows from the fact that if Z becomes more dispersed, then $\min\{Z, z^*\}$ also becomes more dispersed (shown in the Appendix). A similar proof does not apply to dispersion of F , because $\min\{Z, z^*\}$ has a mass on z^* and, therefore, never has log-concave density. We provide a direct proof for the second result in the Appendix. *Q.E.D.*

An increase in dispersion of G tends to increase the value of search, because each consumer always has an option not to take an inspected product and, therefore, her ex post search payoff is convex. Consumers would then visit more sellers and acquire more information, which makes their effective values more dispersed. This result is in stark contrast to its counterpart with unobservable prices: Anderson and Renault (1999) found that the equilibrium price initially decreases but eventually increases as the distribution of match values (G) gets scaled up. This difference is precisely due to the difference in price observability: in the same environment as Anderson and Renault (1999) (with a degenerate F and no outside option), if G is degenerate, then the equilibrium price is minimized in our model, because the pricing game reduces to standard Bertrand competition, while it is maximized in their model, because of the Diamond paradox.

Notice that the two conditions in Proposition 4 are not symmetric. This is due to the aforementioned fact that $\min\{Z, z^*\}$ cannot have log-concave density as long as $s > 0$. Economically, whereas more ex ante preference diversity directly increases ex post preference diversity, it may also deter more consumers from searching and learning about their total values, which reduces preference diversity. Proposition 4 provides a sufficient condition under which more ex ante diversity translates into more ex post diversity. Specifically, the condition in the proposition makes F increase, in the sense of first-order stochastic dominance, as F becomes more dispersed. This ensures that more consumers engage in search and, therefore, the latter effect reinforces, rather than inhibits, the former effect.

6.2. Search Costs

It is well-known that if prices are unobservable before search, then market prices tend to rise with search costs.²⁸ Intuitively, an increase in search costs reduces the value of additional search, which strengthens each seller's market power over visiting consumers

²⁸To be precise, the result depends on the shapes of the relevant distributions. Anderson and Renault (1999) considered the case where F is degenerate and showed that the symmetric equilibrium price p^* increases in s if $1 - G$ is log-concave but decreases in s if $1 - G$ is log-convex, assuming that there exists a symmetric pure-strategy market equilibrium in both cases. Haan, Moraga-González, and Petrikaite (2017) provided some sufficient conditions under which p^* rises with s in the duopoly environment with non-degenerate F . In the Supplemental Material (Section D), we provide an example in which F is non-degenerate, $1 - G$ is log-concave, but p^* decreases in s .

and, therefore, leads to higher prices. The following result shows that the assumption of observable prices reverses the result under a standard regularity assumption.

PROPOSITION 5: *In the symmetric environment, both equilibrium price p^* and equilibrium profit $\pi(p^*)$ decrease in s , provided that f is log-concave.*

PROOF: We prove the price result here and relegate a proof for the profit result to the Appendix. For the price result, we show that as s increases, H becomes less dispersed and $H(u_0 + c)$ increases. The result then follows by applying Proposition 3. If s increases, then z^* falls (see equation (1)). This implies that $W = V + \min\{Z, z^*\}$ decreases in the sense of first-order stochastic dominance and, therefore, $H(u_0 + c)$ rises. A decrease of z^* also reduces dispersion of $\min\{Z, z^*\}$: since the distribution function for $\min\{Z, z^*\}$ coincides with $G(z)$ until z^* and then jumps to 1, the corresponding quantile function is given by $\min\{G^{-1}(a), z^*\}$, which becomes weakly flatter at any $a \in (0, 1)$ as z^* decreases. The desired result then follows by letting $X_1 = V$ and $X_2 = \min\{Z, z^*\}$ and applying the mathematical result in the proof of Proposition 4 above. *Q.E.D.*

Intuitively, observable prices directly influence consumer search: the lower price a seller offers, the more consumers visit him. An increase in search costs raises the value of attracting consumers, because they search less and are more likely to purchase from their visit. This intensifies price competition among the sellers and leads to lower prices. Another way to understand the result (in fact, the idea used in the proof above) is through the relationship between search costs and the distribution of consumers' effective values. An increase in s induces consumers to search less. They stop with lower values (thus, H decreases in the sense of first-order stochastic dominance) and earlier (thus, H becomes less dispersed), both of which provide an incentive for the sellers to lower their prices.

This result has been repeatedly established for the duopoly case (Armstrong and Zhou (2011), Shen (2015), Haan, Moraga-González, and Petrikaite (2017)). The fact that the same result holds under different correlation structures proves the robustness of the underlying economic forces. Proposition 5 complements the previous findings not only by showing that the result goes far beyond the duopoly case, but also by providing an alternative approach based on our discrete-choice reformulation (Theorem 1) and new general comparative statics result (Proposition 3).

In order to better understand the difference between observable and unobservable prices, let p_i denote the actual price seller i charges and p_i^e denote the price consumers expect seller i to charge. Assuming that all other sellers advertise and charge p^* and applying the same logic as for Theorem 1, a consumer purchases product i if and only if

$$\min\{v_i + z^* - p_i^e, v_i + z_i - p_i\} > \max_{j \neq i}\{u_0, \max w_j - p^*\}.$$

For expositional clarity, suppose that there is no outside option and let $\tilde{W}_i \equiv \max_{j \neq i} W_j$. Then, demand for seller i is given by

$$D_i(p_i, p_i^e, p^*) = E\left[\int_{\tilde{W}_i - p^* - z^* + p_i^e}^{\infty} (1 - G(\tilde{W}_i - p^* - v_i + p_i)) dF(v_i)\right]. \tag{5}$$

If prices are unobservable, then in the symmetric equilibrium,

$$-\frac{dD_i(p_i, p_i^e, p^*)}{dp_i}\bigg|_{p_i=p_i^e=p^*} = E\left[\int_{\tilde{W}_i - z^*}^{\infty} g(\tilde{W}_i - v_i) dF(v_i)\right],$$

while if prices are observable (in which case $p_i = p_i^e$), then

$$-\frac{dD_i(p_i, p_i, p^*)}{dp_i} \Big|_{p_i=p^*} = E \left[\int_{\tilde{W}_i - z^*}^{\infty} g(\tilde{W}_i - v_i) dF(v_i) + (1 - G(z^*))f(\tilde{W}_i - z^*) \right].$$

In the second equation, the first term represents consumers on the *intensive* margin, who are indifferent between purchasing product i and the best alternative (i.e., $V_i + Z_i = \tilde{W}_i$) among those who visit seller i (i.e., $V_i + z^* \geq \tilde{W}_i$), while the second term captures consumers on the *extensive* margin, who are indifferent between visiting seller i and the best alternative (i.e., $V_i + z^* = \tilde{W}_i$) but will purchase product i if they visit seller i (i.e., $V_i + Z_i \geq \tilde{W}_i$). As shown above, the union of these two groups of consumers coincides with the event $\{W_i = \tilde{W}_i\}$. When prices are unobservable, only the intensive margin is operative and, therefore, each seller faces fewer marginal consumers. It immediately follows that market prices are higher when they are unobservable before search than when they are observable.

The difference in the extensive margin is also responsible for different comparative statics results regarding search costs. With observable prices, $-dD_i/dp_i$ coincides with the event $\{W_i = \tilde{W}_i\}$, leading to $-dD_i/dp_i = \int h(w) dH(w)^{n-1}$. It then follows that the price effect of search costs can be inferred from its clear effect on dispersion of H . With unobservable prices, this is no longer the case, because the absence of the extensive margin breaks the equality between $-dD_i/dp_i$ and the event $\{W_i = \tilde{W}_i\}$. In fact, the result is ambiguous even under common regularity assumptions. An increase in s lowers z^* and, therefore, reduces the measure of marginal consumers given a realization of $\tilde{W}_i$ (the integral term in the expectation operator), which pushes up market prices. Intuitively, an increase in $s_i (= s)$ makes only the consumers who particularly value product i continue to visit seller i , which provides an incentive for the seller to raise her price. However, the distribution of $\tilde{W}_i$ (i.e., the expectation operator) itself changes with s : it falls in the sense of first-order stochastic dominance and becomes less dispersed as s rises. Depending on the shapes of F and G , this effect can increase or decrease the measure of marginal consumers and may even dominate the first effect. This makes the overall effect ambiguous (see footnote 28).

Proposition 5 opens up an interesting possibility that consumer surplus may increase with search costs. An increase in search costs has a direct negative effect on consumer surplus. However, if the sellers lower their prices drastically in response, then overall consumer surplus may rise. In the Supplemental Material (Section E), we provide an example in which this is indeed the case. It arises when the outside option is sufficiently unfavorable and there are sufficiently few sellers. In that case, each seller possesses strong market power and, therefore, charges a sufficiently high price. An increase in search costs induces them to drop their prices quickly, up to the point where the indirect effect outweighs the direct effect and, therefore, consumer surplus increases.

6.3. Pre-Search Information Quality

In our model, consumers search because they have imprecise information about their values for the products. This means that search frictions can also be measured by the extent to which consumers are uncertain about their total values. We now examine the effect of improving pre-search information quality on the equilibrium price p^* .

For tractability and clarity, we restrict attention to a Gaussian learning environment where both F and G are given by normal distributions with mean 0.²⁹ Furthermore, we assume that F has variance α^2 , while G has variance $1 - \alpha^2$, for some $\alpha \in (0, 1)$ (i.e., $V \sim \mathcal{N}(0, \alpha^2)$ and $Z \sim \mathcal{N}(0, 1 - \alpha^2)$). The variances are deliberately chosen so as to ensure that $\widehat{V} = V + Z \sim \mathcal{N}(0, 1)$ for any α , that is, the distribution for consumers' ex post (total) values is independent of α . The parameter α measures the quality of pre-search information: as α increases, consumers' ex post values $\widehat{V} = V + Z$ are influenced more by V and less by Z . We also assume that consumers have no outside option.

We find that, unlike in Propositions 4 and 5, the equilibrium price p^* (and, therefore, equilibrium profit $\pi(p^*) = (p^* - c)/n$ as well) may or may not increase as pre-search information quality improves. In particular, if the number of sellers is sufficiently large, then the equilibrium price p^* increases in α .³⁰

PROPOSITION 6: *In the symmetric environment where $V \sim \mathcal{N}(0, \alpha^2)$, $Z \sim \mathcal{N}(0, 1 - \alpha^2)$, and $u_0 = -\infty$, for each α that satisfies Assumption 1,³¹ there exists an integer $n^*(\alpha)$ such that p^* increases in α if and only if $n \geq n^*(\alpha)$.*

Recall that market prices increase when H becomes more dispersed (Proposition 3). Importantly, the result depends only on H , not separately on F and G . This suggests that an increase in α has two opposing effects on p^* . On the one hand, it compresses G , which, as shown in Proposition 4, tends to decrease p^* . On the other hand, it spreads out F , which tends to increase dispersion of H and, therefore, drive up p^* .

In order to understand why the result depends on n , notice that, due to the specific truncation structure of $W = V + \min\{Z, z^*\}$, the relative impact of F on H grows larger as w increases. Meanwhile, consumers' maximal effective values are higher when there are more sellers. Together, these imply that the measure of marginal consumers becomes more sensitive to the behavior of F as n increases. Since F becomes more dispersed, while G becomes less dispersed, in α , p^* increases in α if and only if n is sufficiently large.

7. CONCLUSION

We analyze an oligopoly model in which each consumer sequentially searches for the best product based on partial product information and advertised (i.e., fully observable) prices. Unlike in some recent related studies, we do not restrict attention to a symmetric duopoly environment: we allow for any number of sellers and seller heterogeneity. This generalization is due to our eventual purchase theorem, which provides a simple static condition that fully summarizes consumers' sequential search outcomes and allows us to reformulate the pricing game among the sellers as a familiar discrete-choice problem. Exploiting this reformulation, we provide sufficient conditions under which there exists a unique market equilibrium, which is in pure strategies. In addition, by developing and utilizing a new general comparative statics result for discrete-choice models, we obtain

²⁹This Gaussian environment corresponds to the web search model in Choi and Smith (2016), who provided several comparative statics results for optimal sequential search behavior.

³⁰This result is similar to a result on the revenue effect of disclosing more information in auctions. Specifically, Ganuza and Penalva (2010) showed that providing more information, in the sense of dispersive order of conditional expected values, always lowers the seller's revenue if there are only two bidders but increases the seller's revenue if there are sufficiently many bidders. Despite several modeling differences, both results suggest that the intensity of competition is an important determinant for the effects of information provision.

³¹For example, if $s = 0.2$, then Assumption 1 holds (i.e., H is log-concave) when $\alpha \geq 0.313$.

some new and more general comparative statics results regarding preference diversity and search frictions.

Many interesting questions remain open. To name a few, we assume that all sellers are fully committed to their advertised prices. However, hidden fees, in various forms, are prevalent in reality. How does their potential presence affect consumer behavior and sellers' pricing incentives? We consider the case where each seller sells only one product, but it is the exception rather than the rule. How should a multi-product seller price and/or position his products? Should the seller choose an identical price, or introduce different prices, for ex ante symmetric products? If the products are asymmetric, which product should the seller make prominent and how? Finally, we assume that pre-search information quality is exogenously given. However, more information can be provided by a seller's own advertisement or by a platform provider's market design. How much pre-search information are they willing to provide? Does competition necessarily induce the sellers to reveal more information?³²

Our theoretical analysis has some broad implications for empirical research. Simultaneous search (à la Stigler (1961), Chade and Smith (2006)) models situations in which each inspection takes a significant amount of time but the decision must be made within a certain time frame (e.g., college admissions). It is not appropriate for many consumer search problems, because inspection is often straightforward for consumer products. However, it is believed to be more tractable than sequential search and, therefore, has been adopted by various empirical studies (see, e.g., Moraga-González, Sándor, and Wildenbeest (2015), Pires (2017)). Our eventual purchase theorem significantly improves the tractability of sequential search, thereby enhancing its applicability. In addition, it implies that existing theoretical and econometric results on discrete-choice models can be used to analyze consumer search markets. We think that this opens up numerous research opportunities, including econometric identification problems (e.g., what distortions arise if one estimates a discrete-choice model ignoring search frictions?) and revisiting previous empirical studies (e.g., comparing simultaneous search with sequential search).

APPENDIX: OMITTED PROOFS

PROOF OF COROLLARY 1: Let $X_i(\mathbf{v}) \equiv \max\{u_0, \max_{j \neq i} v_j + \min\{Z_j, z_j^*\} - p_j\}$ for each $i = 1, \dots, n$ and $X_0(\mathbf{v}) \equiv \max_i v_i + \min\{Z_i, z_i^*\} - p_i$. By Theorem 1, the consumer eventually purchases product i when $v_i + \min\{Z_i, z_i^*\} - p_i > X_i$. In addition, by the proof logic of Theorem 1, she pays search costs s_i whenever $v_i + z_i^* - p_i > X_i$. Finally, by the definition of z_i^* (equation (1)), $s_i = E[\max\{0, Z_i - z_i^*\}]$. Combining these facts, the consumer's expected payoff from shopping can be written and arranged as follows:

$$\begin{aligned} & \mathbb{1}_{\{X_0 < u_0\}} u_0 + \sum_{i=1}^n E[\mathbb{1}_{\{X_i < v_i + \min\{Z_i, z_i^*\} - p_i\}} (v_i + Z_i - p_i) - \mathbb{1}_{\{X_i < v_i + z_i^* - p_i\}} s_i] \\ &= \mathbb{1}_{\{X_0 < u_0\}} u_0 \\ &+ \sum_{i=1}^n E[\mathbb{1}_{\{X_i < v_i + \min\{Z_i, z_i^*\} - p_i\}} (v_i + Z_i - p_i) - \mathbb{1}_{\{X_i < v_i + z_i^* - p_i\}} \max\{0, Z_i - z_i^*\}] \end{aligned}$$

³²See Ellison (2005) and Anderson and Renault (2006) for some important contributions and Armstrong (2017) and the references therein for some recent developments.

$$\begin{aligned}
&= \mathbb{1}_{\{X_0 < u_0\}} u_0 \\
&\quad + \sum_{i=1}^n E \left[\mathbb{1}_{\{X_i < v_i + \min\{Z_i, z_i^*\} - p_i\}} (v_i + Z_i - p_i) - \mathbb{1}_{\{X_i < v_i + \min\{Z_i, z_i^*\} - p_i\}} \max\{0, Z_i - z_i^*\} \right] \\
&= \mathbb{1}_{\{X_0 < u_0\}} u_0 + \sum_{i=1}^n E \left[\mathbb{1}_{\{X_i < v_i + \min\{Z_i, z_i^*\} - p_i\}} (v_i + Z_i - p_i - \max\{0, Z_i - z_i^*\}) \right] \\
&= \mathbb{1}_{\{X_0 < u_0\}} u_0 + \sum_{i=1}^n E \left[\mathbb{1}_{\{X_i < v_i + \min\{Z_i, z_i^*\} - p_i\}} (v_i + \min\{Z_i, z_i^*\} - p_i) \right] \\
&= E \left[\max \left\{ u_0, \max_i v_i + \min\{Z_i, z_i^*\} - p_i \right\} \right],
\end{aligned}$$

where $\mathbb{1}$ denotes the indicator function.

Q.E.D.

PROOF OF PROPOSITION 2: (1) The survival function $1 - H_i$ is log-concave. Equation (2) can be rewritten as

$$1 - H_i(w_i) = \int_{\underline{z}_i}^{\bar{z}_i} (1 - F_i(w_i - \min\{z_i, z_i^*\})) g_i(z_i) dz_i. \quad (6)$$

Since $-w_i + \min\{z_i, z_i^*\}$ is concave in (w_i, z_i) and $1 - F_i(-x)$ is increasing and log-concave (due to log-concavity of $1 - F_i(x)$) in x , the composite function $1 - F_i(-(-w_i + \min\{z_i, z_i^*\}))$ is log-concave in (w_i, z_i) . Since $g_i(z_i)$ is also log-concave and log-concavity is preserved under multiplication, the product $(1 - F_i(w_i - \min\{z_i, z_i^*\})) g_i(z_i)$ is log-concave in (w_i, z_i) . The desired result then follows from Prékopa's theorem, which states that if the integrand is log-concave, then the integral is also log-concave (see, e.g., Caplin and Nalebuff (1991), Choi and Smith (2016), for a formal statement of the theorem and its uses in related contexts).

(2) If the variance of V_i is sufficiently large, then H_i is also log-concave.

We provide a basic argument here, relegating a complete (long) proof to the Supplemental Material (Section B). Given V_i and Z_i , let $V_i^\sigma \equiv \sigma V_i$ and $W_i^\sigma \equiv V_i^\sigma + \min\{Z_i, z_i^*\}$. In addition, let F_i^σ and H_i^σ denote the distribution functions for V_i^σ and W_i^σ , respectively. By the definition of V_i^σ ,

$$F_i^\sigma(v_i^\sigma) = F_i\left(\frac{v_i^\sigma}{\sigma}\right), \quad f_i^\sigma(v_i^\sigma) = \frac{f_i(v_i^\sigma/\sigma)}{\sigma}, \quad \text{and} \quad (f_i^\sigma)'(v_i^\sigma) = \frac{f_i'(v_i^\sigma/\sigma)}{\sigma^2}.$$

Fixing $F_i^\sigma(v_i^\sigma) = r$ at a quantile, or equivalently $v_i^\sigma = \sigma F_i^{-1}(r)$, as σ explodes, f_i^σ converges to 0 at rate $1/\sigma$, and $(f_i^\sigma)'$ does at rate $1/\sigma^2$. Intuitively, as F_i^σ becomes more spread out, its density f_i^σ falls to 0 and the slope of the density converges to 0 even faster. Exploiting these differences in the converge rates, it follows that given $r \in [0, 1]$, as σ tends to infinity,

$$H_i^\sigma(w_i^\sigma) \rightarrow F_i(v_i), \quad \sigma h_i^\sigma(w_i^\sigma) \rightarrow f_i(v_i), \quad \text{and} \quad \sigma^2 (h_i^\sigma)'(w_i^\sigma) \rightarrow [1 - G_i(z_i^*)] f_i'(v_i),$$

where $w_i^\sigma \equiv (F_i^\sigma)^{-1}(r) + z_i^*$ and $v_i \equiv F_i^{-1}(r)$ (see the Supplemental Material for details). It follows that if σ is sufficiently large, then

$$\text{sign} \left[\frac{(h_i^\sigma)'(w_i^\sigma)}{h_i^\sigma(w_i^\sigma)} - \frac{h_i^\sigma(w_i^\sigma)}{H_i^\sigma(w_i^\sigma)} \right] = \text{sign} \left[\frac{(1 - G_i(z_i^*)) f_i'(v_i)}{f_i(v_i)} - \frac{f_i(v_i)}{F_i(v_i)} \right].$$

Since $(\log H_i^\sigma)'' = ((h_i^\sigma)'H_i^\sigma - (h_i^\sigma)^2)/(H_i^\sigma)^2$, H_i^σ is log-concave if $(h_i^\sigma)' / h_i^\sigma - h_i^\sigma / H_i^\sigma < 0$ for all w_i^σ . Given w_i^σ , this inequality holds when σ is large because the right-hand side of the displayed equation is negative by log-concavity of F_i . In the Supplemental Material, we prove that there exists $\bar{\sigma}$ such that if $\sigma > \bar{\sigma}$, then the right-hand side is negative for all w_i^σ .

(3) If $f_i'(u_0 + c_i - z_i^*) \leq 0$, then H_i is log-concave above $w_i > \max\{u_0 + c_i, \underline{w}_i\}$.

Log-concavity of f_i implies that there exists v_i^* such that $f_i'(v_i) \leq 0$ if and only if $v_i \geq v_i^*$. If $f_i'(u_0 + c_i - z_i^*) \leq 0$, then it must be that $u_0 + c_i - z_i^* \geq v_i^*$ and, therefore, $f_i(v_i)$ decreases in v_i whenever $v_i \geq u_0 + c_i - z_i^* \geq v_i^*$. Differentiating equation (6) with respect to w_i , then

$$h_i(w_i) \equiv H_i'(w_i) = \int_{\underline{z}_i}^{\bar{z}_i} f_i(w_i - \min\{z_i, z_i^*\}) g_i(z_i) dz_i.$$

Suppose $w_i > u_0 + c_i$. If w_i rises, then $f_i(w_i - \min\{z_i, z_i^*\})$ falls for any $z_i \in [\underline{z}_i, \bar{z}_i]$, because

$$w_i - \min\{z_i, z_i^*\} \geq w_i - z_i^* > u_0 + c_i - z_i^*.$$

It follows that $h_i(w_i)$ also decreases in w_i as long as $w_i > u_0 + c_i$. It is then automatic that $h_i(w_i)/H_i(w_i)$ falls in w_i above $w_i > u_0 + c_i$. Q.E.D.

PROOF OF PROPOSITION 3: We prove that the right-hand side of equation (4) rises in p^* . Let $D_i(p_i, p^*, u_0)$ denote demand for seller i when all other sellers choose p^* and consumers' outside option is given by u_0 . It suffices to show the following cross-derivative is positive:

$$\begin{aligned} \frac{d}{dp^*} \left(- \frac{\partial \log(D_i(p_i, p^*, u_0))}{\partial p_i} \Big|_{p_i=p^*} \right) &= \frac{d}{dp^*} \left(- \frac{\partial \log(D_i(p_i + u_0, p^* + u_0, 0))}{\partial p_i} \Big|_{p_i=p^*} \right) \\ &= \frac{d}{du_0} \left(- \frac{\partial \log(D_i(p_i + u_0, p^* + u_0, 0))}{\partial p_i} \Big|_{p_i=p^*} \right) \\ &= - \frac{\partial^2 \log(D_i(p_i, p^*, u_0))}{\partial p_i \partial u_0} \Big|_{p_i=p^*} \geq 0. \end{aligned}$$

Note that the first and the third equations use the fact that $D_i(p_i, p^*, u_0) = D_i(p_i + u_0, p^* + u_0, 0)$. The inequality holds because D_i is log-submodular in (p_i, u_0) by Theorem 1 in Quint (2014). Q.E.D.

PROOF OF PROPOSITION 4: (1) If Z becomes more dispersed, then $\min\{Z, z^*\}$ also becomes more dispersed.

Notice that the quantile function for $\min\{Z, z^*\}$ is given by $\min\{G^{-1}(a), z^*\}$. It suffices to show that its slope increases at any $a \in (0, 1)$. For $a < G(z^*)$, the result is immediate from $\min\{G^{-1}(a), z^*\} = G^{-1}(a)$. For $a \geq G(z^*)$, the result follows from the fact that $G(z^*)$ rises as G becomes more dispersed: rewriting equation (1) with $b^* = G(z^*)$ and $b = G(z)$ yields $s = \int_{b^*}^1 (1 - b) \partial G^{-1}(b) / \partial b db$. If G becomes more dispersed ($\partial G^{-1}(b) / \partial b$ rises), the integrand rises, and thus the lower support b^* must rise in order to maintain the equation.³³

³³This argument is borrowed from Choi and Smith (2016).

(2) Fix V_0 and consider V_1 that is more dispersed than V_0 . For each $i = 0, 1$, let F_i denote the distribution function for V_i . Assume that both f_0 and f_1 are decreasing above $v \geq u_0 + c - z^*$ and the lower bound of their support is given by $\underline{v} \in (-\infty, u_0 + c - z^*]$. Define the quantile $\underline{a}_0 \equiv \Pr\{V_0 + \min\{Z, z^*\} \leq u_0 + c - z^*\}$. In the following, we show that the quantile function of $V_1 + \min\{Z, z^*\}$ is steeper than that of $V_0 + \min\{Z, z^*\}$ for all quantiles above $\underline{a}_0$.

For $t \in [0, 1]$, let V_t be the random variable whose quantile function is given by $F^{-1}(a, t) \equiv (1 - t)F_0^{-1}(a) + tF_1^{-1}(a)$. It is clear that V_t grows more dispersed in t . Since V_1 dominates V_0 in the sense of first-order stochastic dominance (Theorem 3.B.13 in Shaked and Shanthikumar (2007)), V_t also rises, in the sense of first-order stochastic dominance, as t rises. Let $F(v, t)$ and $f(v, t)$ denote the distribution function and the density function of V_t , respectively. Given t and $\underline{b}_0 \equiv F_0(u_0 + c - z^*)$, since V_1 dominates V_0 in the first-order stochastic dominance sense and f_0 and f_1 have decreasing density, $f_v(v, t) \leq 0$ for any $v \geq F^{-1}(\underline{b}_0, t)$.

Let $W_t \equiv V_t + \min\{Z, z^*\}$, and denote by $H(w, t)$ and $H^{-1}(a, t)$ its distribution function and quantile function, respectively. To prove that $\partial H^{-1}(a, t)/\partial a$ rises in t at all $a \in [\underline{a}_0, 1]$, it suffices to show that $H_t(w, t)/h(w, t)$ falls monotonically in w for $w \geq H^{-1}(\underline{a}_0, t)$, because given $a = H(w)$, $\partial[\partial H^{-1}(a, t)/\partial a]/\partial t = \partial[\partial H^{-1}(a, t)/\partial t]/\partial a = -\partial[H_t(w, t)/h(w, t)]/\partial a$ has the same sign as $-\partial[H_t(w, t)/h(w, t)]/\partial w$. Notice that since $u_0 + c - z^* > \underline{v}$, $w \geq \underline{v} + z^*$ whenever $w \geq u_0 + c$ (i.e., whenever w lies in the relevant region). Given $w \geq \underline{v} + z^*$, by equation (2), $H_t(w, t)/h(w, t)$ can be written as

$$\begin{aligned} \frac{H_t(w, t)}{h(w, t)} &= \frac{\int_{\underline{z}}^{\bar{z}} F_t(w - \min\{z, z^*\}, t)g(z) dz}{\int_{\underline{z}}^{\bar{z}} f(w - \min\{z, z^*\}, t)g(z) dz} \\ &= \frac{\int_{\underline{z}-w}^{\bar{z}-w} \frac{F_t(\max\{-r, w - z^*\}, t)}{f(\max\{-r, w - z^*\}, t)} f(\max\{-r, w - z^*\}, t)g(r + w) dr}{\int_{\underline{z}-w}^{\bar{z}-w} f(\max\{-r, w - z^*\}, t)g(r + w) dr} \\ &= E\left[\frac{F_t(\max\{-R, w - z^*\}, t)}{f(\max\{-R, w - z^*\}, t)}\right]. \end{aligned} \quad (7)$$

The second equation is due to a change of variables with $r = z - w$. The random variable R in the last equation has density $f(\max\{-r, w - z^*\}, t)g(r + w)$ and support $(\underline{z} - w, \bar{z} - w)$. The right-hand side falls in w for two reasons. First, as explained above with H , $F_t(v, t)/f(v, t)$ falling in v is equivalent to $F(v, t)$ growing more dispersed in t . So the ratio inside the expectation operator falls in w for any given R . Second, R falls in the first-order stochastic dominance sense in w because (i) the support of R shifts to the left in w and (ii) the density of R is log-submodular in (r, w) by log-concavity of g and by $f_v(w - z^*, t) \leq 0$ for $w \geq H^{-1}(\underline{a}_0, t)$. To see the inequality, recall that $f_v(w - z^*, t) \leq 0$ for $w - z^* \geq F^{-1}(\underline{b}_0, t)$. One can show $H^{-1}(\underline{a}_0, t) \geq F^{-1}(\underline{b}_0, t) + z^*$ for each $t \in [0, 1]$: For $t = 0$, $H^{-1}(\underline{a}_0, 0) = u_0 + c$ and $F^{-1}(\underline{b}_0, 0) = u_0 + c - z^*$ by the def-

inition of $\underline{a}_0$ and $\underline{b}_0$, and thus the inequality binds. For $t > 0$, the inequality holds as $dH^{-1}(\underline{a}_0, t)/dt \geq dF^{-1}(\underline{b}_0, t)/dt$ whenever the inequality binds.³⁴ *Q.E.D.*

PROOF OF THE PROFIT RESULT IN PROPOSITION 5: Let $\mathbf{p}^* = (p_i^*, p_{-i}^*)$ denote the equilibrium price vector. In addition, in a slight abuse of notation, let $D_i(\mathbf{p}^*, z^*)$ denote the equilibrium demand for seller i given $\mathbf{p}^*$ and z^* , and $\pi_i(\mathbf{p}^*, z^*)$ the corresponding profit (i.e., $\pi_i(\mathbf{p}^*, z^*) = (p_i^* - c)D_i(\mathbf{p}^*, z^*)$). By Theorem 1, $D_i(\mathbf{p}^*, z^*) = \Pr\{V_i + \min\{Z_i, z^*\} - p_i^* > X_i\}$ where $X_i \equiv \max\{u_0, \max_{j \neq i} V_j + \min\{Z_j, z^*\} - p_j^*\}$. Let $d\pi_i(\mathbf{p}^*, z^*)/ds$ represent the effect of a marginal increase in s on each seller's equilibrium profit. Then,

$$\frac{d\pi_i(\mathbf{p}^*, z^*)}{ds} = \frac{\partial p_i^*}{\partial s} \frac{\partial \pi_i(\mathbf{p}, z^*)}{\partial p_i} \Big|_{\mathbf{p}=\mathbf{p}^*} + \frac{\partial p_{-i}^*}{\partial s} \frac{\partial \pi_i(\mathbf{p}, z^*)}{\partial p_{-i}} \Big|_{\mathbf{p}=\mathbf{p}^*} + \frac{\partial z^*}{\partial s} \frac{\partial \pi_i(\mathbf{p}, z^*)}{\partial z^*} \Big|_{\mathbf{p}=\mathbf{p}^*}.$$

Each term represents the marginal effect of own price, that of the other sellers' prices, and that of consumer search behavior, respectively. In equilibrium, the first term is 0 by the envelope theorem ($\partial \pi_i(\mathbf{p}, z^*)/\partial p_i = 0$ at $\mathbf{p} = \mathbf{p}^*$). The second term is negative because $\partial p^*/\partial s \leq 0$, as shown above, and $\partial \pi_i(\mathbf{p}, z^*)/\partial p_{-i} \geq 0$, as the products are imperfect substitutes one another. The last term is negative because $\partial z^*/\partial s < 0$ (equation (1)) and $\partial \pi_i(\mathbf{p}, z^*)/\partial z^* \geq 0$: the latter inequality is due to the fact that an increase in z^* raises H in the sense of first-order stochastic dominance, induces fewer consumers to take the outside option and, therefore, raises $D_i(\mathbf{p}^*, z^*)$. Overall, it is necessarily the case that $d\pi(\mathbf{p}^*, z^*)/ds \leq 0$. *Q.E.D.*

PROOF OF PROPOSITION 6: Given that there is no outside option, the equilibrium price p^* is given by

$$\frac{1}{p^* - c} = n \int_{\underline{w}}^{\infty} h(w) dH(w)^{n-1} = n \int_0^1 h(H^{-1}(a)) da^{n-1},$$

where the second equation changes variable $a = H(w)$. By the implicit function theorem,

$$\frac{\partial p^*}{\partial \alpha} = -(p^* - c)^2 n \int_0^1 \frac{\partial h(H^{-1}(a))}{\partial \alpha} da^{n-1}.$$

The desired result follows by letting $\gamma(a) = \partial h(H^{-1}(a))/\partial \alpha$ and applying the next two results. Lemma 1 shows that $\gamma(a)$ is positive and then negative as a rises. Lemma 2 uses Lemma 1 to show that $\int_0^1 \gamma(a) da^n < 0$ if and only if n is large. *Q.E.D.*

LEMMA 1: *There exists $w^* (< \infty)$ such that the slope of $H^{-1}(a)$ decreases in α if and only if $a > H(w^*)$.*

The complete proof is long and is in the Supplemental Material (Section F). Intuitively, if w is large, then the shape of H is mostly determined by F . Since F becomes more dispersed in α , H also does so for w large. If w is small, then H is affected by all three V , Z , and z^* . The effects of the first two cancel each other out, because $V + Z \sim N(0, 1)$.

³⁴By the implicit function theorem, $dH^{-1}(\underline{a}_0, t)/dt = -H_t(w, t)/h(w, t)$ where $w = H^{-1}(\underline{a}_0, t)$, and similarly $dF^{-1}(\underline{b}_0, t)/dt = -F_t(w, t)/f(w, t)$ where $w = F^{-1}(\underline{b}_0, t)$. If $H^{-1}(\underline{a}_0, t) = F^{-1}(\underline{b}_0, t) + z^*$, then $dH^{-1}(\underline{a}_0, t)/dt \geq dF^{-1}(\underline{b}_0, t)/dt$ by equation (7) and the fact that $F_t(w, t)/f(w, t)$ falls in w .

The last effect through z^* , however, makes H less dispersed, because z^* decreases in α (see equation (1) and the proof of Proposition 5).

LEMMA 2: For any real-valued function $\gamma : \mathcal{R} \rightarrow \mathcal{R}$, if $\int_0^1 \gamma(a) da^n \leq 0$ and there exists a' such that $\gamma(a) < 0$ if and only if $a > a'$, then

$$\int_0^1 \gamma(a) da^{n+1} = \frac{n+1}{n} \int_0^1 \gamma(a)a da^n \leq 0.$$

PROOF: The result immediately follows from the fact that a is positive and strictly increasing and, therefore, assigns more weight to the negative portion of $\gamma(a)$ in the integral (see Karlin and Rubin (1955) for a formal proof of this logic). Q.E.D.

REFERENCES

- ANDERSON, S. P., AND R. RENAULT (1999): "Pricing, Product Diversity, and Search Costs: A Bertrand–Chamberlin–Diamond Model," *RAND Journal of Economics*, 30 (4), 719–735. [1262,1271]
- (2000): "Consumer Information and Firm Pricing: Negative Externalities From Improved Information," *International Economic Review*, 41 (3), 721–742. [1260]
- (2006): "Advertising Content," *American Economic Review*, 96 (1), 93–113. [1275]
- (2017): "Search Direction: Position Externalities and Position Bias," Report. [1262]
- ARMSTRONG, M. (2017): "Ordered Consumer Search," *Journal of the European Economic Association*, 15 (5), 989–1024. [1258,1260,1261,1265,1275]
- ARMSTRONG, M., AND J. VICKERS (2015): "Which Demand Systems Can Be Generated by Discrete Choice?" *Journal of Economic Theory*, 158, 293–307. [1258,1261]
- ARMSTRONG, M., AND J. ZHOU (2011): "Paying for Prominence," *Economic Journal*, 121 (556), 368–395. [1258, 1259,1262,1272]
- ATHEY, S., AND G. ELLISON (2011): "Position Auctions With Consumer Search," *Quarterly Journal of Economics*, 126 (3), 1213–1270. [1262]
- BAGNOLI, M., AND T. BERGSTROM (2005): "Log-Concave Probability and Its Applications," *Economic Theory*, 26, 445–469. [1267]
- BAYE, M. R., AND J. MORGAN (2001): "Information Gatekeepers on the Internet and the Competitiveness of Homogeneous Product Markets," *American Economic Review*, 91 (3), 454–474. [1262]
- BHASKAR, V., AND E. HOPKINS (2016): "Marriage as a Rat Race: Noisy Premarital Investments With Assortative Matching," *Journal of Political Economy*, 124 (4), 992–1045. [1261]
- CAPLIN, A., AND B. NALEBUFF (1991): "Aggregation and Imperfect Competition: On the Existence of Equilibrium," *Econometrica*, 59 (1), 25–59. [1276]
- CHADE, H., AND L. SMITH (2006): "Simultaneous Search," *Econometrica*, 74 (5), 1293–1307. [1275]
- CHEN, Y., AND C. HE (2011): "Paid Placement: Advertising and Search on the Internet," *Economic Journal*, 121 (556), 309–328. [1262]
- CHEN, Y., AND M. H. RIORDAN (2008): "Price-Increasing Competition," *RAND Journal of Economics*, 39 (4), 1042–1058. [1268]
- CHOI, M., A. Y. DAI, AND K. KIM (2018): "Supplement to 'Consumer Search and Price Competition'," *Econometrica Supplemental Material*, 86, <https://doi.org/10.3982/ECTA14837>. [1259]
- CHOI, M., AND L. SMITH (2016): "Optimal Sequential Search Among Alternatives," Working Paper 2858097, SSRN. [1274,1276,1277]
- DE LOS SANTOS, B., A. HORTAÇSU, AND M. R. WILDENBEEST (2012): "Testing Models of Consumer Search Using Data on Web Browsing and Purchasing Behavior," *American Economic Review*, 102 (6), 2955–2980. [1262]
- DINERSTEIN, M., L. EINAV, J. LEVIN, AND N. SUNDARESAN (2017): "Consumer Price Search and Platform Design in Internet Commerce," *American Economic Review* (forthcoming). [1262]
- DOVAL, L. (2018): "Whether or Not to Open Pandora's Box," *Journal of Economic Theory*, 175, 127–158. [1263]
- DUBOIS, P., AND H. PERRONE (2015): "Price Dispersion and Information Frictions: Evidence From Supermarket Purchases," Discussion Paper DP10906, CEPR. [1260]
- ELLISON, G. (2005): "A Model of Add-on Pricing," *Quarterly Journal of Economics*, 120 (2), 585–637. [1275]

- GABAIX, X., D. LAIBSON, D. LI, H. LI, S. RESNICK, AND C. G. DE VRIES (2016): "The Impact of Competition on Prices With Numerous Firms," *Journal of Economic Theory*, 165, 1–24. Available at <http://www.sciencedirect.com/science/article/pii/S0022053116300035>. [1268,1270]
- GANUZA, J.-J., AND J. S. PENALVA (2010): "Signal Orderings Based on Dispersion and the Supply of Private Information in Auctions," *Econometrica*, 78 (3), 1007–1030. [1261,1274]
- GARCIA, D., J. HONDA, AND M. JANSSEN (2017): "The Double Diamond Paradox," *American Economic Journal: Microeconomics*, 9 (3), 63–99. [1261]
- HAAN, M., J. L. MORAGA-GONZÁLEZ, AND V. PETRIKAITE (2017): "A Model of Directed Consumer Search," Discussion Paper DP11955, CEPR. [1258-1260,1262,1268,1271,1272]
- KARLIN, S., AND H. RUBIN (1955): "The Theory of Decision Procedures for Distributions With Monotone Likelihood Ratio," Applied Mathematics and Statistics Laboratory. Available at <http://books.google.co.uk/books?id=3JT2NwAACAAJ>. [1280]
- KIM, J. B., P. ALBUQUERQUE, AND B. J. BRONNENBERG (2010): "Online Demand Under Limited Consumer Search," *Marketing Science*, 29 (6), 1001–1023. [1262]
- KLEINBERG, R., B. WAGGONER, AND E. G. WEYL (2017): "Descending Price Optimally Coordinates Search," Working Paper 2753858, SSRN. [1258,1261,1265]
- KOULAYEV, S. (2014): "Search for Differentiated Products: Identification and Estimation," *RAND Journal of Economics*, 45 (3), 553–575. [1260]
- MORAGA-GONZÁLEZ, J. L., Z. SÁNDOR, AND M. R. WILDENBEEST (2015): "Consumer Search and Prices in the Automobile Market," IESE Research Papers D/1123, IESE Business School. Available at <https://ideas.repec.org/p/ebg/iesewp/d-1123.html>. [1260,1275]
- (2017): "Prices and Heterogeneous Search Costs," *RAND Journal of Economics*, 48 (1), 125–146. [1261]
- PERLOFF, J. M., AND S. C. SALOP (1985): "Equilibrium With Product Differentiation," *Review of Economic Studies*, 52 (1), 107–120. [1259,1262,1269]
- PIRES, T. (2017): "Measuring the Effects of Search Costs on Equilibrium Prices and Profits," *International Journal of Industrial Organization* (forthcoming). [1260,1275]
- QUINT, D. (2014): "Imperfect Competition With Complements and Substitutes," *Journal of Economic Theory*, 152, 266–290. [1262,1266,1268,1277]
- SHAKED, M., AND J. G. SHANTHIKUMAR (2007): *Stochastic Orders*. Springer Science & Business Media. [1269-1271,1278]
- SHELEGIA, S., AND D. GARCIA (2018): "Consumer Search With Observational Learning," *RAND Journal of Economics*, 49 (1), 224–253. [1261]
- SHEN, J. (2015): "Ex-ante Preference in a Consumer Search Market," Working Paper 2826011, SSRN. [1258, 1259,1262,1272]
- STIGLER, G. J. (1961): "The Economics of Information," *Journal of Political Economy*, 69 (3), 213–225. [1275]
- VIVES, X. (2005): "Complementarities and Games: New Developments," *Journal of Economic Literature*, 43 (2), 437–479. [1266]
- WEITZMAN, M. L. (1979): "Optimal Search for the Best Alternative," *Econometrica*, 47 (3), 641–654. [1257, 1258,1262-1264]
- WOLINSKY, A. (1986): "True Monopolistic Competition as a Result of Imperfect Information," *Quarterly Journal of Economics*, 101 (3), 493–511. [1261,1262]
- WRIGHT, R., P. KIRCHER, B. JULIEN, AND V. GUERRIERI (2017): "Directed Search: A Guided Tour," Working Paper 23884, NBER. [1261]
- ZHOU, J. (2014): "Multiproduct Search and the Joint Search Effect," *American Economic Review*, 104 (9), 2918–2939. [1261,1262]
- (2017): "Competitive Bundling," *Econometrica*, 85 (1), 145–172. Available at <http://dx.doi.org/10.3982/ECTA14251>. [1259,1261,1270]

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